

Todd H. Albert, Ph.D.

Executive Brief

Engineering Executive | Complex Systems Rebuilder | SaaS, AI, Cloud, and Organizational Transformation

Todd Albert rebuilds complex systems.

Across science, education, startups, and enterprise software, his career has followed a consistent pattern: enter environments where complexity has outgrown structure, find the signal inside the disorder, and rebuild the system so it becomes clearer, stronger, and more capable of scaling.

Todd is a technology executive, former NASA Graduate Research Fellow, educator, founder, hands-on architect, and author. He brings more than 30 years of software experience, 15+ years of leadership, and 20+ years of academic and scientific depth to organizations facing difficult technical and operational transitions.

He currently serves as Vice President of Engineering at CYPHER Learning, a global SaaS learning platform serving millions of users across multiple production environments. There, he leads engineering, DevOps, QA, and integrations for a 32+ person distributed organization with a \$4.4 million budget. His work includes rebuilding engineering operating rhythms across planning, delivery, QA, cloud operations, and executive reporting; improving accountability and execution predictability; advancing AI, automation, security, compliance, and data visibility; and presenting technology strategy, roadmap progress, risk, and financial metrics to private-equity board stakeholders.

Under his leadership, CYPHER has reduced cloud infrastructure spend by more than \$1.2 million annually while improving platform resilience, performance, and operational visibility. That outcome reflects the center of Todd's leadership style: he does not treat cost, architecture, delivery, and team health as separate problems. He treats them as one system.

Before CYPHER, Todd served as Vice President of Application Development at Site Impact, where he inherited a stalled \$4.3 million AdTech platform initiative after two and a half years of missed deadlines. He simplified the architecture, reduced technical ambiguity, reset execution practices, and helped restore organizational confidence in the platform. He also replatformed core systems for multi-region resilience, reducing projected infrastructure spend by approximately \$600,000 annually, while building engineering, DevOps, and SRE practices from the ground up.

As CTO of Edgy Nerd, Todd founded and led a product engineering studio delivering SaaS, fintech, AI, analytics, payment, and cloud-native platforms across AWS, Azure, and Google Cloud. He architected secure and resilient systems for regulated and high-growth markets, managed engineering teams, client delivery, vendors, budgets, and P&L, and engineered a nationally recognized AI diarization application combining speech processing, automation, and production-grade cloud architecture.

Earlier, as Head Software Engineer for MyPhoto / Polaroid Photo Bar, Todd rebuilt the company's custom e-commerce and fulfillment platform in under six months, modernizing it from PHP to Node.js, React, and cloud-native infrastructure. The transformation increased conversion from 2% to 10%, cut bounce rate by 50%, reduced production time by 66%, and increased productivity 10x through AI, computer vision, Python, React Native, and Electron automation. He also invented an "instant upload" experience that removed friction from the customer ordering flow.

At MadValorem, Todd rebuilt a stalled \$1.2 million real estate data platform from the ground up, migrating the core system from .NET to a LAMP architecture. He solved a mapping and geospatial search problem that had blocked a 12-engineer team and built deterministic content-generation systems for more than 600,000 properties, developments, neighborhoods, cities, counties, and states. His technical and strategic impact earned him a 33% equity stake in the company.

Todd's builder instinct extends beyond software products. He founded Boca Code to strengthen South Florida's software engineering talent pipeline and has led or advised technical delivery across SaaS, fintech, AI, AR, e-commerce, and education platforms, including work connected to PBS Kids, AccuWeather, Ally Bank, Sprint, and early-stage SaaS ventures.

His foundation is unusually deep. Before becoming an engineering executive, Todd was a NASA Graduate Research Fellow conducting glacier, remote sensing, and climate research. He published peer-reviewed work on glaciers and climate systems, contributed to *Global Land Ice Measurement from Space*, authored an award-recognized encyclopedia entry, evaluated scientific grants for NASA's Mission to Planet Earth, and later founded and led the first Meteorology BA/BS program in Tennessee. He taught more than 2,500 students and helped grow a university major sevenfold, preserving department viability.

That scientific and academic foundation still shapes his executive approach. Todd is not a chaos manager. He is a systems thinker. When a platform is fragile, a delivery process is unreliable, or a team is underperforming, he looks beneath symptoms for the structural cause: unclear ownership, brittle architecture, weak feedback loops, missing standards, misaligned incentives, or decisions that no longer match the scale of the business.

His leadership philosophy is direct: create the conditions where people can do the best work of their careers. In practice, that means building trust, clarity, operating discipline, and technical standards without resorting to blame. He engages at the level of strategy, architecture, execution, and people — because in real organizations, those systems are inseparable.

Todd is also the author of *Out of the Fish Tank* and a regular writer on engineering, product, AI, and leadership. His writing extends the same reconstructive discipline that defines his executive work: finding structure inside complexity and turning scattered information into something coherent enough to hold.

Todd is best understood as an executive systems rebuilder: a leader who works where software, teams, architecture, strategy, and story intersect — and who helps complex organizations become legible, scalable, and alive again.

Selected Executive Proof Points

- Leads engineering, DevOps, QA, and integrations for a global SaaS learning platform serving millions of users across multiple production environments.
- Manages a 32+ person distributed organization and \$4.4 million budget.
- Reduced cloud infrastructure spend by more than \$1.2 million annually while improving resilience, performance, and operational visibility.
- Presents technology strategy, roadmap progress, risk, and financial metrics to private-equity board stakeholders.
- Rescued a stalled \$4.3 million AdTech platform initiative after 2.5 years of missed deadlines.
- Reduced projected infrastructure spend by approximately \$600,000 annually through multi-region platform modernization.
- Built engineering, DevOps, SRE, CI/CD, observability, incident readiness, and infrastructure-as-code practices from the ground up.
- Founded and led a product engineering studio delivering SaaS, fintech, AI, analytics, payment, and cloud-native platforms across AWS, Azure, and Google Cloud.
- Engineered and scaled a nationally recognized AI diarization application using speech processing, automation, and production-grade cloud architecture.
- Rebuilt Polaroid Photo Bar's e-commerce and fulfillment platform in under six months.
- Increased conversion from 2% to 10%, cut bounce rate by 50%, reduced production time by 66%, and increased productivity 10x through AI and computer vision automation.
- Rebuilt a stalled \$1.2 million real estate data platform and solved a geospatial search problem that had blocked a 12-engineer team.
- Built structured content-generation systems for more than 600,000 properties, developments, neighborhoods, cities, counties, and states.
- Founded Boca Code to strengthen South Florida's engineering talent pipeline.
- NASA Graduate Research Fellow with peer-reviewed research in glaciers, remote sensing, and climate systems.
- Founded and led the first Meteorology BA/BS program in Tennessee.
- Taught more than 2,500 students and helped grow a university major sevenfold.